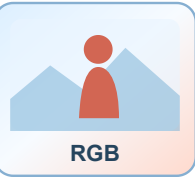


HTL-ReID: Hierarchical Token Learning for RGB-NIR-TIR Re-Identification

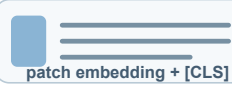
Shared encoding, mask-guided interaction, local representation, and independent heterogeneous fusion

MULTI-MODAL INPUT

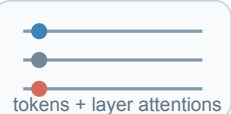


Shared ViT Backbone

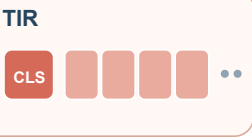
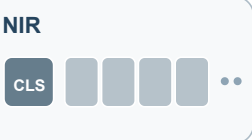
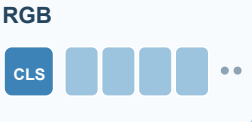
same parameters for all modalities



Transformer blocks
MSA + MLP
x L shared layers



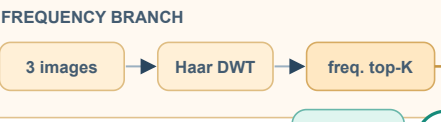
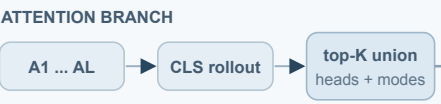
BACKBONE TOKENS



complete RGB / NIR / TIR tokens

HS Hierarchical Selection

one shared hard patch mask



shared M_HS

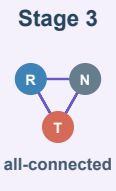
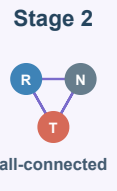
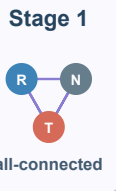
rollout mask OR frequency mask = complete HS selection

selected patch features

ACI Adaptive Cross-modal Interaction

full tokens, constrained by M_HS

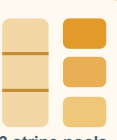
Each target CLS attends to both other modalities
sample-adaptive routing + gated residual update



ACI descriptor (3d)

PLR Part-aware Local Representation

independent local descriptor



concat
x 3 modes
PLR (9d)

Final Descriptor

per-branch L2 normalization

ACI	ACI descriptor	1.0
CLS	Original CLS	1.1
PLR	PLR descriptor	0
DHF	DHF descriptor	1.58

weighted concatenation
then final L2 normalization

Frozen E045 inference
PLR weight = 0 (branch retained)

DHF Dynamic Heterogeneous Fusion

seven parallel routes from complete backbone tokens

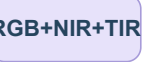
3 SINGLE-MODAL



3 BI-MODAL



1 ALL-MODAL



route-token attention + residual experts
one expert per heterogeneous route

sample-adaptive gate
4 heads over 7 routes

DHF
descriptor 7d

complete backbone-token flow

HS-selected patch flow

HS mask constraint

d = ViT token dimension

All module labels are paper-facing names.